

Ecuación Cuadrática: $\overbrace{ax^2}^{\text{término cuadrático}} + \overbrace{bx}^{\text{término lineal}} + \overbrace{c}^{\text{término independiente}} = 0$

$$ax^2 + bx + c = a(x - x_1)(x - x_2)$$

a : coeficiente del término cuadrático
 b : coeficiente del término lineal
 c : término independiente

$$x_1, x_2 = \frac{-b \pm \sqrt{b^2 - 4 \cdot a \cdot c}}{2 \cdot a}$$

1. Halle el conjunto solución, completando cuadrados:

i. $x^2 = 3x + 28$

ii. $x(2x + 15) = -18$

iii. $x^2 + 13 = 4x$

iii) $x^2 + 13 = 4x$

1° ordenar:

$$x^2 - 4x + 13 = 0$$

2° $a=1$:

3° pasar c al lado derecho:

$$x^2 - 4x = -13$$

4° Sumar $m \cdot a \cdot m$:

$$\left(\frac{b}{2 \cdot a}\right)^2$$

$$x^2 - 4x + 2^2 = -13 + 2^2$$

5° factorizar el trinomio

$$(x - 2)^2 = -13 + 4 = -9$$

6° Aplicar raíz en ambos lados

$$(x - 2)^2 = -9$$

7° Simplificar →

$$\sqrt{(x - 2)^2} = \sqrt{-9}$$

8° Resolver el módulo →

$$|x - 2| = 3i$$

$$\begin{cases} x - 2 = 3i \\ x = 3i + 2 \end{cases}$$

$$x = 2 + 3i$$

$$\begin{cases} -(x - 2) = 3i \\ x - 2 = -3i \\ x = -3i + 2 \end{cases}$$

$$x = 2 - 3i$$

$$\left(\frac{b}{2 \cdot a}\right)^2 = \left(\frac{-4}{2 \cdot 1}\right)^2 = \left(\frac{-4}{2}\right)^2 = (-2)^2 = 4$$

$$(x - 2)^2 =$$

$$\sqrt{-9} = \sqrt{9} \cdot (\sqrt{-1})^i = 3 \cdot i$$

No tiene soluciones o raíces en los números reales, tiene raíces complejas

3. Halle el conjunto solución de cada ecuación:

i. $2x^2 = 32$; ii. $x^2 - 25 = 0$

iii. $-5x^2 = 10$

v. $4x = 3x^2 - 2x$

vii. $\frac{2x}{x-3} + \frac{x+1}{2x+6} = \frac{3}{2x-6}$

ii. $x^2 + 9 = 0$

iv. $x^2 = x$

vi. $-x + 10x^2 - 6x = 2x^2 + x$

viii. $\frac{x-2}{x+1} + \frac{4x}{x-1} = \frac{8}{x^2-1}$

$$ii) x^2 - 25 = 0$$

$$x^2 = 25$$

$$x = \pm \sqrt{25}$$

$$\boxed{x_1 = 5}$$

$$\boxed{x = -5}$$

$$a=1$$

$$b=0$$

$$c=-25$$

$$x_{1,2} = \frac{0 \pm \sqrt{0^2 - 4 \cdot 1 \cdot (-25)}}{2 \cdot 1} = \frac{\pm \sqrt{100}}{2} = \frac{\pm 10}{2}$$

$$x_1 = \frac{10}{2} = 5$$

$$x_2 = \frac{-10}{2} = -5$$

$$\boxed{x_1 = 5}$$

$$\boxed{x_2 = -5}$$

otra forma Diferencia de Cuadrados

$$x^2 - 25 = 0$$

$$x^2 - 5^2 = 0$$

$$(x-5) \cdot (x+5) = 0$$

$$\downarrow \quad \downarrow$$

$$\boxed{x=5} \quad \boxed{x=-5}$$

$$iv) x^2 = x$$

$$x^2 - x = 0$$

$$x \cdot (x-1) = 0$$

$$\downarrow$$

$$\boxed{x=0} \quad \boxed{x=1}$$

$$\left\{ \begin{array}{l} a=1 \\ b=-1 \\ c=0 \end{array} \right. \quad x_{1,2} = \frac{1 \pm \sqrt{(-1)^2 - 4 \cdot 1 \cdot 0}}{2 \cdot 1} = \frac{1 \pm \sqrt{1-0}}{2} = \frac{1 \pm \sqrt{1}}{2}$$

$$= \frac{1 \pm 1}{2} \rightarrow x_1 = \frac{1+1}{2} = \frac{2}{2} = 1 \quad \boxed{x_1 = 1}$$

$$x_2 = \frac{1-1}{2} = \frac{0}{2} = 0 \quad \boxed{x_2 = 0}$$

EC. cuadrática
Expresado
como
factores lineales

$$x^2 - x = x \cdot (x-1)$$

$$ax^2 + bx + c = a \cdot (x-x_1)(x-x_2)$$

$$vi. -x + 10x^2 - 6x = 2x^2 + x$$

$$-x + 10x^2 - 6x - 2x^2 - x = 0$$

$$8x^2 - 8x = 0$$

$$8x \cdot (x-1) = 0$$

$$\downarrow \quad \downarrow$$

$$\boxed{x=0} \quad \boxed{x=1}$$

$$\frac{\cancel{8}x^2}{\cancel{8}x} = x^1$$

$$x \neq 3 \vee x \neq -3$$

$$vii. \frac{2x}{x-3} + \frac{x+1}{2x+6} = \frac{3}{2x-6}$$

$$\frac{2x}{x-3} + \frac{x+1}{2x+6} - \frac{3}{2x-6} = 0$$

$$2x+6 = 2 \cdot (x+3)$$

$$2x-6 = 2 \cdot (x-3)$$

$$\frac{2x}{x-3} + \frac{x+1}{2x+6} - \frac{3}{2x-6} = 0$$

$$2x+6 = 2(x+3)$$

$$2x-6 = 2(x-3)$$

$$\frac{2x}{x-3} + \frac{x+1}{2(x+3)} - \frac{3}{2(x-3)} = \frac{2x \cdot 2(x+3) + (x+1) \cdot (x-3) - 3(x+3)}{(x-3) \cdot 2 \cdot (x+3)} = 0$$

$$\frac{2x \cdot 2(x+3) + (x+1)(x-3) - 3(x+3)}{2 \cdot (x-3)(x+3)} = 0$$

$$12x - 3x + x - 3x =$$

$$4x(x+3) + x^2 - 3x + x - 3 - 3x - 9 = 0 \cdot 2(x-3)(x+3) = 0$$

$$4x^2 + 12x + x^2 - 3x + x - 3 - 3x - 9 = 0$$

$$5x^2 + 7x - 12 = 0$$

$$a=5$$

$$b=7$$

$$c=-12$$

$$x_{1,2} = \frac{-7 \pm \sqrt{7^2 - 4 \cdot 5 \cdot (-12)}}{2 \cdot 5} = \frac{-7 \pm \sqrt{49 + 240}}{10} = \frac{-7 \pm \sqrt{289}}{10}$$

$$= \frac{-7 \pm 17}{10} \rightarrow \begin{cases} x_1 = \frac{-7+17}{10} = \frac{10}{10} = 1 & \boxed{x_1=1} \\ x_2 = \frac{-7-17}{10} = \frac{-24}{10} = \frac{-12}{5} & \boxed{x_2 = -\frac{12}{5}} \end{cases}$$

ANÁLISIS DEL DISCRIMINANTE (Δ)

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4 \cdot a \cdot c}}{2 \cdot a}$$

$$\Delta = b^2 - 4 \cdot a \cdot c$$

- $\Delta > 0 \Rightarrow$ 2 soluciones reales y distintas
- $\Delta = 0 \Rightarrow$ 2 soluciones reales e iguales
- $\Delta < 0 \Rightarrow$ 2 soluciones complejas.

$$\begin{aligned} x_1 &= \frac{-b + \sqrt{b^2 - 4 \cdot a \cdot c}}{2 \cdot a} & x_2 &= \frac{-b - \sqrt{b^2 - 4 \cdot a \cdot c}}{2 \cdot a} \\ x_1 &= \frac{-b+0}{2 \cdot a} = \frac{-b}{2 \cdot a} & x_2 &= \frac{-b-0}{2 \cdot a} = \frac{-b}{2 \cdot a} \end{aligned}$$

4. Determine "k" para que la ecuación $3x^2 - 6x + k = 0$ tenga una raíz doble.

$$\Delta = b^2 - 4 \cdot a \cdot c$$

$$3x^2 - 6x + k = 0$$

$$a=3$$

$$b=-6$$

$$c=k$$

$$\Delta = 0$$

$$\rightarrow \Delta = (-6)^2 - 4 \cdot 3 \cdot k = 0$$

$$36 - 12k = 0$$

$$-12k = -36$$

$$k = \frac{-36}{-12} = 3$$

Verificación

$$3x^2 - 6x + 3 = 0$$

$$a=3$$

$$b=-6$$

$$c=3$$

$$\begin{aligned} x_{1,2} &= \frac{+6 \pm \sqrt{(-6)^2 - 4 \cdot 3 \cdot 3}}{2 \cdot 3} = \frac{6 \pm \sqrt{36 - 36}}{6} \\ &= \frac{6 \pm \sqrt{0}}{6} \rightarrow x_1 = \frac{6+0}{6} = \frac{6}{6} = 1 \end{aligned}$$

$$K = \frac{-36}{-12} = 3$$

$$\boxed{K=3}$$

$$= \frac{6 \pm \sqrt{0}}{6}$$

$$\rightarrow x_1 = \frac{6+0}{6} = \frac{6}{6} = 1$$

$$\rightarrow x_2 = \frac{6-0}{6} = \frac{6}{6} = 1$$